

One firewall setting is breaking every class

Teachers' **video** freezes while their **voice** stays clear. We found why. The fix is a router setting — it costs **nothing** and takes minutes. This page explains what to ask for, and how to check it worked.

4 September 2026 · replaces the 3 September version (the cause turned out to be different) · based on measurements from our own database

Technical words are marked with a small orange numberⁿ — they are all explained in plain English in the **Glossary at the end**. You only really need **1, 2 and 4**.

WHAT WE NEED FROM YOU

Please allow outbound UDP¹ on the office internet

Ask whoever manages the office router / firewall⁴ to stop blocking outgoing UDP traffic. That is the whole request.

Cost: ₱0

No new hardware

No plan upgrade

Takes minutes

COPY THIS TO YOUR IT PERSON OR ISP

Our video-call software (WebRTC) needs to send **outbound UDP**. Right now it is being blocked, so every call falls back to TCP and the video freezes.

Please allow **outbound UDP** from the office network. Ideally allow outbound UDP to any port. If that is not possible, at minimum please allow outbound UDP to:

- `turn.cloudflare.com` — **UDP 3478** and **UDP 53**
- `stun.cloudflare.com` — **UDP 3478** and **UDP 53**
- `stun.l.google.com` — **UDP 19302**

We do not need any *inbound*⁴ ports¹⁰ opened. Nothing needs to be exposed to the internet.

If the router already allows UDP — then the blocking is happening at the internet provider.

Please ask the ISP: *"Is UDP blocked or rate-limited on this line? Are we behind CGNAT⁸?"* Their answer tells us where to go next.

1 Why this makes video freeze but not voice

The internet has two ways to send things. Think of them as **courier** and **registered mail**.

UDP¹ — the courier

If a package is late, it is thrown away and the next one goes through. Video looks a little rougher for a second, then continues. **This is what video calls are built for.**

TCP² — registered mail

Every package must arrive, **in order**. If one is late, everything behind it waits in line. Video stops dead, then jumps. **This is what we are stuck on now.**

Voice survives registered mail because it is tiny — about 30 kbps, and the browser can paper over small gaps. Video is roughly **40× bigger**. One late packet stalls the whole picture, then a burst arrives. That is exactly the “freeze... jump... freeze” teachers describe.

So the problem is not your internet speed. We measured round-trip times⁷ of 113–324 ms during the broken calls — that is a perfectly normal, healthy connection. The traffic is simply being forced down the wrong lane.

2 What we measured

From our own database, 4 September 2026. Every video call now records which route it took.

What we looked at	What we found	What it should be
Calls that connected directly (peer to peer) ⁹	0 out of 46	Most of them
Calls forced through a relay server ⁵	46 out of 46 — 100%	Only a few
Relay connections using UDP (the courier)	0	All of them
Relay connections using TCP (registered mail)	35 out of 35 — 100%	0
Round-trip time during the broken calls	113–324 ms	Normal — not the problem
Video freezes	4–9 per minute	Near zero
Audio gaps the browser had to fill in	4–12%	Under 2%

We checked our own side first. Our server does hand out UDP routes — we verified it on 4 September. So the UDP lane is offered; something on the network path is throwing those packets away.

This affected **6 different people**, not one bad laptop. Roughly ten teacher accounts and two managers log in from the same office IP address, so one router setting explains all of it.

3 What we already fixed on our side

We are not asking you to carry this alone. Since 3 September we shipped:

- **Stopped the flicker.** When the line got bad, our software switched to audio-only and then tried to come back after only 8 seconds — over and over. That is why the picture kept appearing and vanishing. It now waits 32 seconds of a genuinely stable line before restoring video.
- **Stopped teachers' laptops recording.** Every browser in the room was making its own recording, which used the same upload and the same CPU as the live video. Teachers no longer record automatically.
- **Lower start on slow lines.** Calls now begin at a lower quality when the line is known to be slow, instead of flooding it and then backing off.
- **We can now see the route.** Every call records whether it went direct or through a relay, and over UDP or TCP. That is how we found this — and it is how we will confirm your fix worked.
- **A backup path (server-relayed video, an SFU¹¹).** Built and ready, switched off. It helps on connections that can never be opened up — but it is **not** a substitute for this fix. It is also WebRTC, so where UDP is blocked it falls back to TCP as well.

4 How we will know it worked

You do not have to test anything or report back with numbers. **Just tell us the date the change was made.** We will read it from our own records.

1. You change the firewall setting and tell us the date.
2. Teachers restart their browser and run one normal class.
3. We look at our route records for that class.

Success looks like this: calls start showing up as **direct** instead of relay, or at least as relay over **UDP** instead of TCP — and video freezes drop from 4–9 per minute toward zero.

If nothing changes, that is still useful — it tells us the blocking is at the ISP, not the office router, and we take it up with them next.

5 Please send this back to us

Question	Your answer
Who manages the office router / firewall?	-----
Is outbound UDP currently blocked?	Yes / No / Don't know
Date the change was made	-----
If it could not be changed — why?	-----
Internet provider and plan name	-----
Are teachers on Wi-Fi or wired cable?	-----

The last two are for later. **The first four are what matter this week.**

6 If UDP truly cannot be opened

Then we move down this list, cheapest first. **Please do not spend money on these until the firewall answer comes back** — a bigger plan will not help if the packets¹² are still being blocked.

Option	Cost	What it would do
Wired cable instead of Wi-Fi for teachers	₱0	Removes one source of loss. Worth doing regardless.
Turn on our server-relay path	₱0 for now	Shortens the stalled leg. Helps, but does not remove the TCP problem.
A second, separate line for classes only	Monthly fee	Bypasses the office firewall entirely. This is the reliable fallback.
Business plan with a public IP	Higher monthly fee	Removes CGNAT if that turns out to be the cause.

7 In one paragraph

Teachers' voices are fine and their video freezes because the office connection is forcing video down a lane it was never designed for. The line is fast enough — we measured it. One firewall setting, at no cost, is very likely to fix it. **Please ask your IT person or ISP to allow outbound UDP, and tell us the date it was done.** We will confirm from our side.

Glossary — the technical words, in plain English

Follow the small orange numbers from the text above. You do not need all of these — **1, 2 and 4** are enough to make the request.

1 · UDP

The **“throw it away if it’s late” way of sending data**. Video calls and online games use it, because being *on time* matters more than being complete. A little roughness is fine; stopping is not. **This is the lane that is currently blocked.**

2 · TCP

The **“everything arrives, in order” way**. Web pages and downloads use it, because being complete matters more than being fast. It is the worst possible choice for live video: one late packet and everything behind it waits.

3 · WebRTC

The **standard technology** browsers use to send live video and audio to each other. Our classroom is built on it; Zoom and Google Meet use the same family. **It is designed around UDP.**

4 · Firewall · outbound vs inbound

A firewall is the **gatekeeper that decides which traffic may pass** (usually built into the router). **Outbound** is traffic leaving your office; **inbound** is traffic coming in from the internet.  We are only asking about **outbound**, so this does not weaken your security — nothing new becomes reachable from outside.

5 · TURN relay

A server that **forwards video on your behalf** when two people cannot reach each other directly. It is meant to be an emergency exit used occasionally. Right now **100% of calls are going through it.**

6 · STUN

A small **“what does my address look like from outside?” lookup server**. A direct connection cannot even be attempted without asking it first — and it also uses UDP. **So when UDP is blocked, direct connections never even get started.**

7 · Round-trip time (RTT)

How long a signal takes to **go and come back**. Korea ↔ Philippines is normally 100–300 ms. We measured 113–324 ms during the broken calls — **completely normal**, which is why “the line is too slow” does not explain what teachers are seeing.

8 · CGNAT

When the internet provider makes **many customers share one internet address**. It saves addresses, but it means nobody outside can pinpoint you — which **makes direct connections much harder**. A plan with a public IP removes it.

9 · Direct connection (peer to peer)

The teacher’s computer and the student’s computer talking **straight to each other**, with no server in between. Fastest and best. We currently have **0 out of 46**.

10 · Port

If an internet address is a **building**, a port is the **room number**. **UDP 3478** means “please let deliveries reach room 3478 in that building.” The three lines in the request are the rooms our classroom uses.

11 · SFU (server-relayed video)

Everyone sends to a **nearby server**, which forwards it on — the structure Zoom and Google Meet use. We have built one and left it switched off.  **It is not a replacement for this request** — it is also WebRTC, so where UDP is blocked it falls back to TCP in exactly the same way.

12 · Packet

One **small piece of data**. A single second of video is split into hundreds of them. “Packets are being dropped” means those pieces are disappearing on the way.